

Planning

decomposition

1. Create the plant and zombie classes
2. Create the peashooter, repeater, and zombie objects
3. Add attacking, taking damage, and movement
4. Create the turn loop
5. Add target selection and win conditions

pattern recognition

The plants attack the zombie each turn if they are alive

The zombie checks its distance each turn

The zombie moves closer when its distance is greater than 0

The zombie attacks a living plant when its distance reaches 0

Health and win conditions are checked after attacks

abstraction

The plants require name, health, and damage

The zombie requires name, health, damage, and distance

Both characters require the methods for attacking and taking damage

The zombie requires a method for moving

algorithm design

1. Create the two plants and one zombie
2. Start the first turn
3. Attack the zombie by the living plants
4. If the zombie is defeated, then the plants won and the game is finished
5. If the zombie is still alive, then move closer to the plants or attack a living plant
6. If the both plants are defeated, then the zombie won and the game is finished
7. Otherwise, start the next turn

uml

```
+-----+
| Plant |
+-----+
| - name |
| - health |
| - damage |
+-----+
```

```

| + attack(zombie) |
| + take_damage(amount) |
+-----+
|
| attacks
v
+-----+
| Zombie |
+-----+
| - name |
| - health |
| - damage |
| - distance |
+-----+
| + move() |
| + attack(plant) |
| + take_damage(amount) |
+-----+

```

Plant attacks Zombie
 Zombie attacks Plant
 Zombie moves toward Plant